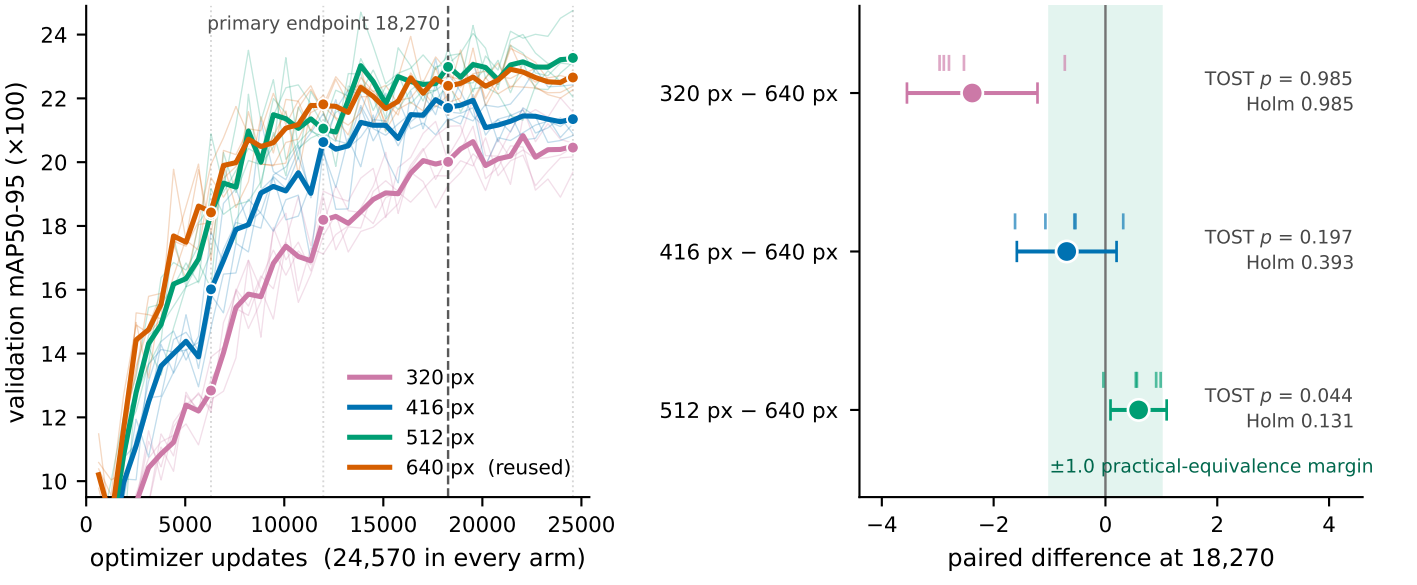
(a) 512 px matches 640; 320 px does not
primary endpoint 18,270
validation mAP50-95 (×100)
optimizer updates (24,570 in every arm)
320 px
416 px
512 px
640 px (reused)
(b) Only 512 px reaches the ±1.0 margin
320 px − 640 px
416 px − 640 px
512 px − 640 px
TOST p = 0.985
Holm 0.985
TOST p = 0.197
Holm 0.393
TOST p = 0.044
Holm 0.131
±1.0 practical-equivalence margin
paired difference at 18,270